A MORPHISM PSEUDO-COHERENT OVER A PSEUDO-COHERENT BASE IS PSEUDO-COHERENT

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 17439-17443 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 17445-17452, the complete author proof (6 body lines).

Editorial note. Navigation stubs: 1 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. BORDERLINE: single-label citation proof; flagged for the Chief.

Editorial note. Statement/proof sentence 'then R -> B is pseudo-coherent' repeats a hypothesis (intended A -> B); retained verbatim, author slip noted, not corrected.

Editorial note. Counted A side only (spaces twin proof incomplete per ruling v3).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. More on Algebra, lemma composition-relative-pseudo-coherent.

Source reference. Original lemma 067D: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 17452–17452. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : X \to Y$ be a morphism of schemes pseudo-coherent over a base scheme S. Then f is pseudo-coherent.*

Proof. This translates into the following algebra result: If $R \to A \to B$ are composable ring maps and $R \to A$, $R \to B$ pseudo-coherent, then $R \to B$ is pseudo-coherent. This follows from More on Algebra, Lemma 067D. □

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.